

Mastering product-market fit: A detailed playbook for AI founders

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Atlas Editors

PMF isn't binary; it's a spectrum. Here's early stage advice along with eight principles to drive traction and reach customers right for your startup.

In the early days of building a company, founders are often told, "You'll know you've achieved product-market fit (PMF) when you *feel* it." But in the AI era, intuition can sometimes be misleading and traditional signals of PMF might be false positives. Many founders will learn the hard way that wildly fast early traction does not always translate into sustainable ARR. AI experimentation budgets are higher than ever, but the threat of churn looms large over many AI app builders.

| PMF isn't binary; it's a spectrum.

That's why SaaS and AI founders in 2025 need to refine their approach to achieving PMF with clearer signals, sharper metrics, and a mindset focused on systematic discovery. While some fundamentals of 'startup physics' remain the same, an AI-first world is also upending many assumptions we've held over the past several decades of SaaS. So what are the new rules of PMF in an AI-first world?

Here is our comprehensive guide to finding PMF in this brave new world—including critical mindsets to embrace, new principles to heed, AI-native signals to watch, and stories from some of today's most impressive AI success stories.

Two critical perspectives to embrace on the road to PMF

The fundamentals of PMF largely remain the same in the AI era: You need to define a really clear ideal customer profile (ICP), solve an urgent pain for this market, and establish positioning that compellingly illustrates why your company is the best to solve it. (And if you'd like a refresher on these fundamentals, see Bessemer Operating Advisor Kim Caldbeck's [6 P framework](#).)

However, much has changed in a tech ecosystem dominated by AI. Before we get down to brass tacks, it's important to embrace the right mental models when thinking about product-market fit in the AI era.

Hitting the PMF bullseye is challenging since it's a moving target

While achieving product-market fit has always been a dynamic process—after all, markets, competitors, and industry landscapes are always in flux—AI intensifies the speed and volatility by an order of magnitude. The potential of AI is vast, but we are still in the earliest innings of AI adoption.

Customers are just beginning to experiment to learn how they want to use AI in their daily life and work. “One of the things that's unique in this moment is that market and buyer preferences and needs are changing at the same time that founders are trying to find PMF,” says Lauri Moore, Partner at Bessemer. “It’s like buying shoes for a growing kid. One day they fit perfectly, and the next they are too snug or their favorite color has changed.”

Now more than ever, founders must stay attuned to frequent changes in customer needs, buyer preferences, and competition from incumbents and new entrants alike.

PMF is a spectrum, not a binary

Founders must recognize that product-market fit is a spectrum, not a binary. There is no elusive “eureka” moment where you can rest easy that you’ve achieved it perfectly. Rather, keep your radar dialed into how your product meets the dynamic needs of your customers.

Product-market fit develops over time and, if your business shows promise, will become a stronger and stronger signal:

Light signal PMF: A handful of early users love the product, but retention is inconsistent.

Moderate signal PMF: You’re seeing pockets of traction—strong usage or early revenue in a segment.

Strong signal PMF: Retention is high, word-of-mouth kicks in, and customers are pulling the product faster than you can deliver.

AI products are often also category creators or first movers in a market. While there may be droves of curious and early advocates, founders must keep in mind that initial positive reception is a light signal of PMF. Initial usage needs to be followed by repeatability with users in a well defined segment. Novelty isn’t the same as value. If users don’t integrate your product into their daily workflows, you don’t have PMF yet.

“I don’t like to declare victory on PMF until a very late stage,” says Adam Fisher, Partner at Bessemer. Celebrate early wins, but remain cautiously optimistic until the signal is strong and unmistakable.

“The startup community sometimes talks about PMF in a way that feels like a summit you climb once,” says Lauri. “But more like a garden you tend daily. With the right materials and efforts, it starts to bloom. But the seasons change — customer needs, competitors, and broader technology trends change — so the work continues.”

Eight principles for achieving PMF

1. Start with a wedge, not a platform

One of the biggest mistakes AI founders make is overbuilding before they start to achieve PMF. AI makes it tempting to go broad, but the best AI startup applications begin by solving one, or a narrower set, of high-pain and high-impact problems by automating workflows. Teams should begin with the wedge and validate the value the initial solution solves for before building out too many features.

PMF isn't about perfection. It's about proving your product solves a real, persistent problem for a real customer with repeatable usage. The shortest path to true product-market fit is to build small and specific ways to automate certain workflows, measure deeply, and iterate on the problem quickly for improved user experience. As Adam Fisher puts it, once you create the initial [spark](#) of product-market fit, you can grow it into a roaring flame. But without creating a true spark, you'll never have a fire to feed.

"The main challenge with PMF in the AI era is that while fast-growing wedge products have become more attainable than ever, they're not nearly as sticky as SaaS products," says Adam. "This is because they're mostly freemium, usage-based and not integrated with company data or existing workflows. So leveraging initial wedge products into something that results in contracted deals and retention should be the most pressing goal."

However, Operating Advisor Allyson Letteri provides an alternative perspective: "Many AI products gain traction quickly because they leverage company data and fit into existing workflows. They deliver value almost instantly, which leads to stronger retention."

In these cases, AI solutions don't need to automate everything, but rather reduce the friction and time it takes to complete a task.

2. Start with a tight, narrowly defined ICP

A common trap AI founders tend to fall into is trying to serve too many customer types too soon. Instead, focus on one ideal customer profile (ICP) at a time. PMF is clearest when you serve a single, tightly defined ICP. Especially in AI, where workflows vary widely across user types, you need concentrated patterns to build strong foundations. Then, you can land and expand within organizations and branch out to adjacent customers or build new acts, based on the needs and user behavior of different cohort groups.

3. Validate use cases, not just ideas

With many horizontal AI products (e.g. ChatGPT, Claude, Perplexity, etc.), it often feels like the sky's the limit in terms of potential use cases. However, brilliant AI ideas aren't worth much unless they solve a real problem for a real set of users. That's why it's critical for AI founders to focus on

delivering an MVP that shows *near* immediate, clear, and undeniable value—even if it’s scrappy. When measuring success, you’ll want to observe actual behavior, time to value, and usage metrics, not just positive feedback.

4. Deliver a demo that showcases the value ASAP

The success of any product demonstration requires showcasing the “wow” moment in the sales process. If a prospect doesn’t see the value during a trial, how are they supposed to trust they’ll gain value from the solution once it’s purchased? Companies that allow prospects to very quickly see how seamlessly their product adds value to their existing workflows—especially without deep integration required—will have much more success driving quick adoption.

“The AI tools that will win their markets will be very plug-and-play to show value,” says Operating Advisor Kim Caldbeck. “Sometimes AI tools require deep integration with customer data and systems. But companies aren’t going to want to do this unless the AI company has already shown traction and has a reputation.” However, increasingly, we’re seeing how new tools and [systems of action](#) make implementation that much easier.

Once you deliver value, focus on designing for repeatable use. Embedding into essential workflows is key, here, and knowing how and *how frequently* your AI product alleviates pain also helps determine success KPIs and strategies to encourage more usage and deeper retention.

5. Ship fast, integrate faster

LLMs evolve weekly. The best AI apps stay model-agnostic, build flexible infrastructure, and update fast to stay ahead of the curve.

“PMF becomes more product-mark-*grit* than in the non-AI era,” says Adam. “Customers often don’t even know what they want, how they will use a product, and who else in the organization is trying a different AI-based product that might supplant this one.” That’s why continuous experimentation and speed are essential.

[Vapi](#), the developer platform for building, testing, and deploying voice AI agents, is a great example of this principle in action. The team unlocked product-market fit during their Y Combinator cohort when they pivoted from earlier iterations to being an AI-driven voice platform. Within six months of this shift, Vapi’s revenue soared to millions of dollars, and its developer base surpassed 100,000. The platform’s flexibility—allowing developers to build custom voice agents for diverse industries—led to rapid adoption and millions of calls handled monthly.

Along with their developer-first approach, usage-based pricing, and ability to scale across diverse verticals, from customer service to healthcare, signaled strong fit and positioned the company well to dominate the voice AI market.

6. Be ruthless about proving ROI—but also appeal to end users

Especially in B2B, AI apps that replace headcount, reduce costs, or drive revenue are easier to sell—and stickier. That's why it's critical to know your value proposition in economic terms and lead with these metrics in your marketing campaigns and sales conversations with C-suite decision makers. However, this approach must also be carefully balanced with messaging that speaks to end users of products, warns Allyson.

"In the buying process, many AI companies make the mistake of alienating the doers who will be using the product day-to-day," she says. "Job elimination obviously doesn't sound great to them. And giving their C-suite full visibility into their daily tasks may feel like overkill. That's why you need to carefully toe the line with your messaging."

Often this looks like different messages for different key members of the buying committee, she advises. "End users often respond well to messages about supercharging their capabilities and eliminating more mundane tasks," she says. Achieving this balance of messaging is not only critical for signing deals, but also long-term retention since it hinges on end users adopting the product and staying happy.

7. Create urgency via strategic messaging and positioning

Since AI is in its early days, most companies face the classic category creator challenge: Before they educate audiences on their products, they must first educate them on the problems they're facing. There is always an entrenched way of doing things that you're up against in your sales and marketing efforts.

"You must first position against the status quo non-AI way of doing things," says Kim. "Then secondarily you need to convince your audience why you are the right AI tool for the job ahead of others in the race."

Allyson suggests that marketing leaders focus on creating urgency via their messaging to shake teams out of their complacency. This often looks like educating on the hidden costs of inefficient solutions, the necessity to future proof, and the very real threat of getting left behind in competitive industries.

8. Unlock viral distribution loops

For AI companies that will ultimately win their markets, unlocking a self-sustaining distribution loop—where current customers draw in new customers—will be critical to achieving market dominance quickly.

This formula was critical for generative media platform [fal.ai](#). For its team, PMF became clear when organic developer adoption and enterprise interest skyrocketed. The company's focus on being the best backend for AI builders—especially for generative media across image, video, and audio created

a viral flywheel. Developers publicly showcased their fal-powered projects, drawing more users without heavy marketing spend.

In just one year, fal.ai's revenue grew 22x, as the team handled 100 million+ inference requests daily and onboarded over 1 million developers including enterprise customers like Canva, Quora, and Perplexity. This speedy and self-perpetuating distribution model helped the team cement its role as critical infrastructure for developers looking for ultra-fast, scalable APIs for AI-powered image, video, and audio generation.

Seven unmistakable signals of product-market fit

Since experimental AI hype can easily be mistaken for *true* product-market fit, it's important to have clear criteria to separate real indicators from the noise.

Here the seven signals to track signs of genuine PMF, as well as what's distinct about tracking this in the AI era:

1. Time to value

For AI tools, since switching costs are low, if it takes too long to show the "magic," churn will follow. That's why you'll want to monitor time to value (TTV) to understand how quickly a user experiences real value. Do users see ROI *immediately* after onboarding or a live demo?

2. Engagement

Monitor your active users (on a daily, weekly, and monthly basis) as well as session length and feature usage depth. Feature usage depth will give you reassurance that your users are taking advantage of your highest-value features. Note that it's common with AI products to see big usage spikes in the initial experimentation phases, but the true test of PMF is if this high level of usage is sustained or deepened over time. Look for signs of habits forming. Is usage increasing, not just in number of users, but in frequency or breadth of use? Are your power users doing more, not just returning?

Another surefire sign of PMF to watch for is if users bypass your UI and integrate into their workflow. If people are creating Zapier workflows, browser extensions, or scripting around your product, you've likely struck a nerve. For AI products, you'll want to track how much of the user's workflow your product automates or enhances. If they're reverting to manual processes, the AI value isn't sticking.

3. Retention

Retention is a classic metric to monitor, but traditional SaaS benchmarks may not apply directly to AI apps. As a hypothetical example, a benchmark of 60–70% at 90 days may still just be a sign of early experimentation with a wide variety of use cases. Instead, measure **retention by use case** to suss out whether or not your customers are using your tools consistently for essential workflows. Finding

extraordinary retention in even one high-value use case is often a stronger signal of product-market fit for AI tools than overall retention, as it identifies exactly where your product is becoming indispensable.

Since many AI projects in 2025 are initially experimental, you'll also want to track second-order engagement. For example, does a user return to work on a new project in your product, not just the first one?

Second-bite usage rate—measuring if users return to the product and repeat similar usage patterns (or net new ones) after completing their first. This serves as a powerful AI-native metric that distinguishes between fleeting experimentation and true product adoption.

For example, [CEO Aravind Srinivas told us at Cloud 100 that Perplexity](#) they time specific tracks cohort analysis and the number of queries people conduct within a specific timeframe; “Initially 80% of users that do one query do another. We wanted to focus on getting this to 100% so two queries can become five queries, and ultimately usage becomes a habit.”

4. Customer feedback

Net promoter score (NPS) is still a powerful indicator of PMF measuring how likely you are to recommend a product to a peer. But there's another inroad to valuable customer feedback — ask your user base, “How would you feel if you could no longer use this product?” If over 40% say they'd be “very disappointed,” you know you're on the right track.

Customer love also has a strong qualitative component—if you catch wind of users spontaneously advocating for your product on Reddit, at events, or on online forums, this is a strong signal of PMF — and even an opportunity to capitalize on organic word-of-mouth marketing.

5. Revenue

For AI products, it's essential to separate experimental ARR from **durable ARR**. If you can separate revenue that's driven by pilots, trials, or novelty spikes from revenue driven by recurring, usage-based, or contractual commitments, you will get a clear picture of how well your product is actually driving value. Durable ARR should ultimately be your North star.

6. Expansion rate

Much like Slack's success hinged on expansion first within teams and then across teams, AI companies should seek to ‘land and expand’ within organizations. For the next generation of AI giants, taking advantage of a viral coefficient for distribution will be the key to rapid success. **Time-to-organization-wide usage** measures how long it takes to go from pilot adoption to organization-wide usage. We can track this by measuring the rate at which new users onboard relative to existing users within an organization.

Similarly, you'll want to monitor use case expansion within teams and organizations. PMF will mean you'll start to see your product shift from novelty use cases to mission-critical ones. AI products often start with fun or lightweight value props. True PMF is reached when the product is no longer optional to someone's job.

7. Positive sales signals

The sales process can be incredibly revealing of whether or not you've reached PMF. Are customers buying fast? Re-upping? Expanding use without persuasion? If your product seems to sell itself, you're onto something. Customer referrals, unprompted word of mouth, and inbound organic growth are material drivers are all important signals to pay attention to. With AI, utility spreads fast—if it's actually useful.

Poll your sales team to understand how much resistance they're facing with prospects—or not. If they report that customers immediately “get it,” and they don't have to over-explain, this is a positive indicator. Similarly, pay attention if buyers says, “I've been waiting for a tool like this” and never want to go back to the old way of doing things.

Sales signals have both a qualitative and quantitative component. Metrics to monitor are **length of sales cycle**, **win rate**, **sales efficiency**, and **ratio of inbound vs. outbound leads**.

Finally, if you're able to develop a repeatable sales motion where you see repeatable buying behavior, from the same type of company, and can use the same process, it's a strong sign of PMF.

Case study: How Brisk Teaching found its sweet spot in the edtech market

[Brisk Teaching](#)'s journey to PMF offers a compelling case study in how to align technological innovation with acute user needs. The tool is an AI-powered Chrome and Microsoft Edge extension designed to streamline workflows for K-12 educators. By addressing painful systemic challenges in education such as teacher burnout, technology fragmentation, and the demand for personalized instruction, the company drove impressive rapid organic adoption across schools and districts.

As of March 2025, less than 18 months post launch, Brisk has reached over 1 million educators across 100+ countries. This growth trajectory, driven primarily by word-of-mouth recommendations and teacher communities, underscored the tool's immediate resonance with educators' daily workflows.

The Brisk team knew they were hitting PMF when educators reported saving over 10 hours a week: time they could put back into teaching, not admin. The team's educator-first mindset is demonstrated by glowing teacher testimonials like: “I was able to grade in 1 hour what would have taken me *all day*... Brisk has given me back my weekend with my family.”

Four PMF lessons AI startups can learn from

Lesson 1: Embed, don't disrupt

When the Brisk team began building, teacher feedback revealed a critical insight: the average teacher juggles 9 different applications daily, and adding another platform only exacerbated their cognitive load. Building a Chrome extension allowed Brisk to embed AI tools directly into teachers' existing workflows (for example, giving feedback within Google Docs or adapting YouTube videos into lesson plans). Furthermore, this positioning reduced onboarding time to under 2 minutes and eliminated the context-switching that made users drop off when trying new applications. District leaders cited Brisk's ability to unify fragmented workflows (e.g., grading, lesson planning, and student feedback) as a key driver of adoption.

Lesson 2: Build advocacy loops

Brisk has continued to cultivate user love through initiatives like Club Brisk, a community of hundreds of educator-advocates who beta-tested features, created instructional resources, and led professional development workshops. The Brisk AI Professional Certification program further entrenched adoption, with teachers earning credentials tied to district continuing education requirements.

This customer love has become a flywheel that extends beyond the digital world. At edtech conferences, teachers advocate for Brisk, with attendees reporting that their peers implore them to try the tool.

Lesson 3: Quantify time savings

Brisk's PMF was reinforced by partnerships with over 2,000 schools and districts. Teachers reported saving more than 10 hours a week, alongside meaningful reductions in cognitive load and improvements in professional satisfaction. These tangible outcomes fueled a groundswell of advocacy and deepened user love.

Lesson 4: Viral adoption flywheel

Brisk's Chrome extension design, embedded directly into tools like Google Docs, YouTube, and PDF viewers made it nearly frictionless to adopt the platform. And since teachers could access AI-powered features without disrupting existing workflows, the Brisk team created a viral adoption loop. For example, educators sharing Brisk-generated lesson plans or feedback tools in online forums was a key contributor to Brisk's exponential growth, with >1 million teachers using the platform by 2025.

Brisk's journey reminds us that product-market fit isn't a single breakthrough, it's the result of many thoughtful choices compounding over time such as relentlessly prioritizing the user, measuring real impact, and embedding deeply in the ecosystems that matter. As the race for AI category leadership accelerates, Brisk offers a powerful blueprint for building enduring products.

A call for new PM/Fit cases studies

In the AI era, product-market fit isn't a binary state, but a dynamic, ongoing process that demands relentless curiosity, rapid iteration, and a willingness to challenge assumptions. The fundamentals—solving a real pain for a clearly defined customer—remain, but the speed and volatility of AI markets require founders to be even more systematic, data-driven, and attuned to shifting user needs.

Don't be fooled by early spikes in experimentation or vanity metrics; true PMF is signaled by sustained engagement, seamless integration into workflows, and customers who can't imagine working without your product. But in any new era of technology comes new standards, expectations, and paths to navigating this ongoing challenge from the builders defining the next cohort of AI giants.

If you're a founder with a story on how your startup found Product/Market fit, share it on LinkedIn and tag Bessemer Venture Partners or email content@bvp.com, if you want to potentially be featured in an upcoming guide or newsletter.

There were many people whose insights contributed to this post including, Bessemer Venture Partners Adam Fisher, Lauri Moore, Kent Bennett, Charles Birnbaum, Investor Maha Malik, including Operating Advisors Allyson Letteri, Kim Caldbeck, among many others.